



Rehabilitation and Return-to-Play Criteria Following Ulnar Collateral Ligament Reconstruction

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Elbow injuries in overhead athletes, particularly ulnar collateral ligament (UCL) injuries to baseball pitchers, continue to increase in number. The reasons for these injuries are multifactorial but typically result from a combination of severe and repetitive valgus elbow stress from pitching large volumes and throwing at peak velocities. These factors and the severity of medial elbow injuries to the UCL complex make it critical to employ a proper criteria-based rehabilitation program in combination with a well-structured set of return-to-play criteria. Together, the rehabilitation process and return-to-play program provide a set of guidelines that help ensure a safe and effective process to aid in returning a throwing athlete back to unrestricted competition as efficiently as possible. This article provides the reader with a 4-phased, criteria-based rehabilitation program for the athlete following UCL reconstruction and a structured set of return-to-play guidelines. The return-to-play progression is a 3-step process that assists in advancing an athlete back to unrestricted competition via a performance, practice, and play sequence. This program uses specific criteria and activities to progress a throwing athlete from high-level performance training, to controlled practice participation and then finally back to play. The program uses not only specific steps but also a combination of testing parameters both subjective and objective to guide the return through the entire rehabilitation process from UCL reconstruction surgery back to unrestricted symptom-free throwing.

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Introduction

Injuries to the elbow joint in the overhead throwing athlete, and especially the baseball pitcher, continue to increase in frequency.^{1,2} There are numerous proposed and documented causes for this increase in incidence.^{1,2} With the rise in elbow injuries, it is important for the clinician to successfully treat and return the individual back to sports safely and reduce their risk

of reinjury. Thus, a thorough and well-structured return-to-play criteria is imperative to ensure a safe return to play.

Overhead throwing athletes subject their elbows to severe and repetitive valgus stress. The ulnar collateral ligament (UCL), particularly the anterior band of the anterior oblique ligament of the UCL complex, which is the primary soft tissue restraint to valgus stress, is commonly injured during the act of throwing.³ Chronic UCL insufficiency results from microscopic tears and attenuation.⁴ Studies have indicated that the valgus stress at the elbow during the acceleration phase of throwing can be as high as 64 N-m, which exceeds the ultimate tensile strength of the UCL.³ Patients report pain and soreness in the medial elbow with throwing, usually in the late cocking or early acceleration phases or with ball release. Athletes often report an episode of sudden onset of pain and “giving way” during throwing. The most common situation is an acute exacerbation of a chronically injured ligament.

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Table 1 Postoperative Rehabilitation After Ulnar Collateral Ligament Reconstruction Using an Autogenous Palmaris Longus Graft

Phase I: immediate postoperative phase (weeks 0-3)

Goals: Protect healing tissue, reduce pain and inflammation, retard muscle atrophy, and protect graft site to allow healing.

(1) Postoperative week 1

- Brace: posterior splint at 90° elbow flexion
- Range of motion (ROM): wrist active range of motion (AROM) extension/flexion immediately postoperative
- Elbow ROM: day 1
- Elbow postoperative compression dressing: 5-7 days
- Wrist (graft site) compression dressing: 7-10 days as needed
- Exercises:
 - Gripping exercises
 - Wrist ROM
 - Shoulder isometrics (no external rotation [ER] of the shoulder)
 - Biceps isometrics
- Cryotherapy: to elbow joint and to graft site at the wrist

(2) Postoperative week 2

- Brace: elbow ROM 25°-100° (gradually increase ROM by 5° extension/10° flexion per week)
- Exercise:
 - Continue all exercises previously listed
 - Elbow ROM in brace (30°-105°)
 - Initiate elbow extension isometrics
 - Continue wrist ROM exercises
 - Initiate light scar mobilization over distal incision (graft)
- Cryotherapy: continue ice to elbow and graft site

(3) Postoperative week 3

- Brace: elbow ROM 15°-115°
- Exercises:
 - Continue all exercises previously listed
 - Elbow ROM in brace
 - Initiate active ROM for wrist and elbow (no resistance)
 - Initiate light wrist flexion and stretching
 - Initiate active ROM for the shoulder:
 - Full can exercises
 - Lateral raises
 - External rotation/internal rotation (ER/IR) exercises with tubing
 - Elbow flexion/extension
- Initiate light scapular strengthening exercises
- May incorporate bicycle workouts for lower extremity strength and endurance

Phase II: intermediate phase (weeks 4-7)

Goals: gradually advance to full ROM; promote healing of repaired tissue; regain and improve muscle strength; restore full function of graft site.

(1) Week 4

- Brace: elbow ROM 0°-125°
- Exercises:
 - Begin light-resistance exercises for the arm (0.5 kg [1 pound]):
 - Wrist curls, extensions, pronation, and supination
 - Elbow extension/flexion

Table 1 (continued)

- Progress shoulder program; emphasize rotator cuff and scapular strengthening
- Initiate shoulder strengthening with light dumbbells

(2) Week 5

- ROM: elbow ROM 0°-135°
- Discontinue brace
- Continue all exercises:
 - Progress all shoulder and upper extremity (UE) exercises (progress weight 0.5 kg [1 pound])

(3) Week 6

- AROM: 0°-145° without brace (ie, full ROM)
- Exercises:
 - Initiate thrower's 10 program
 - Progress elbow strengthening exercises
 - Initiate shoulder external rotation strengthening
 - Progress shoulder program

(4) Week 7

- Exercises:
 - Progress thrower's 10 program (progress weights)
 - Initiate proprioceptive neuromuscular facilitation (PNF) diagonal patterns (light)

Phase III: advanced strengthening phase (weeks 8-14)

Goals: Increase strength, power, and endurance; maintain full elbow ROM; gradually initiate sports/functional activities (specificity).

(1) Week 8

- Exercises:
 - Initiate eccentric elbow flexion/extension
 - Continue shoulder program (thrower's 10 program)
 - Initiate manual resistance diagonal patterns
 - Initiate plyometric exercise program (2-handed plyometrics close to the body only):
 - Chest pass
 - Side throw close to the body
 - Continue calf and hamstring stretching

(2) Week 10

- Exercises:
 - Continue all exercises listed previously
 - Advance plyometrics to 2-handed drills away from the body:
 - Side-to-side throws
 - Soccer throws
 - Side throws

(3) Weeks 12-14

- Continue all exercises
- Initiate strengthening exercises on isotonic machines (if desired):

Table 1 (continued)

• Bench press (seated) • Lat pull down
○ Initiate golf, swimming ○ Initiate interval hitting program
Phase IV: return-to-activity phase (weeks 14-32) Goals: Continue to increase strength, power, and endurance of upper extremity musculature Gradual return to sports/functional activities. (4) Week 14 • Exercise: ○ Continue strengthening program ○ Emphasize elbow and wrist strengthening and flexibility exercises ○ Maintain full elbow ROM ○ Initiate 1-handed plyometric throwing (stationary throws) ○ Initiate 1-handed wall dribble ○ Initiate 1-handed baseball throws into wall (5) Week 16 • Exercises: ○ Initiate interval throwing program (phase I) (long toss program) ○ Continue thrower's 10 program and plyometrics ○ Continue stretching before and after throwing (6) Week 22-24 • Exercises: ○ Progress to phase II throwing (once phase I has been successfully completed) (7) Week 30-32 • Exercises: ○ Gradually progress to competitive throwing/sports ○ Functional activities

Associated ulnar neuropathy is quite common,⁵ and patients may report pain or numbness in the ulnar nerve distribution. Associated pathology, such as loose bodies, osteophytes, and a flexion contracture, can also produce symptoms.

Pitchers sustain most baseball-related injuries, accounting for 61% of the total injuries in major league baseball, and 72% of the injuries suffered by pitchers are to the throwing shoulder and elbow.¹ Elbow injuries account for between 22% and 26% of all major league baseball pitching injuries.^{2,6} Conte et al¹ reported that 25% of all major league baseball pitchers and 14% of all minor league pitchers on the active roster have undergone UCL surgery. As the incidence of UCL injury is increasing so are the number of surgeries performed. Hodgins et al⁷ reported a 193% increase in UCL reconstruction during the past 10 years in the state of New York. The greatest risk factors that contribute to suffering a pitching injury to the UCL in youth baseball pitchers are pitching when fatigued, to a great

volume, improper mechanics, and repetitive throwing at maximal effort. The greatest risk for a professional major league baseball pitcher to sustain a UCL injury is pitching at peak velocity; thus, throwing at high velocity with a large amount of pitches has been shown to increase UCL injury risk.⁸

Treatment for UCL Deficiency

The initial treatment of UCL injury is generally nonoperative⁹ and consists of rest from sport, anti-inflammatory medications, and rehabilitation. One report indicates that about half of these patients can be treated successfully without surgery and are able to return to the same level of athletic activity.¹⁰ Surgery is indicated if symptoms recur after an appropriate trial of nonoperative treatment. Surgical treatment of a UCL injury



Figure 1 Low load-long duration stretching to improve elbow extension. A low intensity stretch is applied for 10-12 minutes. Note that the elbow is pronated and the shoulder is internally rotated to lock the humerus and prevent compensation. (Color version of the figure available online.)

should also be considered with acute ruptures in high-level throwing athletes, with significant chronic instability, and after debridement of UCL calcification.

Primary repair of acute ruptures of the UCL had been advocated for years; however, this has changed because of the overall consistently better results obtained with ligamentous reconstruction.^{9,11} Although most tears are midsubstance (87%), some are avulsion injuries, and all have been considered amenable to suture repair.¹² This was the treatment of

Table 2 Thrower's 10 Exercise Program

Diagonal pattern D2 extension
Diagonal pattern D2 flexion
External rotation at 0° abduction
Internal rotation at 0° abduction
Shoulder abduction to 90°
Scaption, external rotation ("full cans")
Side-lying external rotation
Prone horizontal abduction
Prone horizontal abduction (full external rotation, 100° abduction)
Prone rowing
Prone rowing into external rotation
Press-ups
Push-ups
Elbow flexion
Elbow extension
Wrist extension
Wrist flexion
Wrist supination
Wrist pronation



Figure 2 Proprioceptive neuromuscular facilitation (PNF) exercises to the elbow to enhance elbow stability. (Color version of the figure available online.)

choice until 1992, when Conway et al¹² reviewed their results with repair and reconstruction. They recommended reconstruction, finding that the overhead athletes performed significantly worse following suture repair compared with reconstruction. They noted that repair should be considered only if the tear was a proximal avulsion, if the procedure was performed soon after injury, if the rest of the ligament was undamaged, and if no ulnar nerve symptoms were present. A newly developed procedure is referred to as the UCL repair with internal brace. This surgical technique uses a synthetic material placed within the injured UCL and secured at the humerus and ulna with anchors. This UCL repair technique is reserved for partial sprains with a significant amount of healthy tissue that is amenable to repair.¹³

UCL reconstruction with a free tendon graft is performed for acute rupture in overhead sports athletes and for chronic UCL instability and elbow pain with UCL instability. The procedure has evolved considerably over the years.¹⁴⁻¹⁶ Several graft options are available. A palmaris longus autograft is most often used, but other sources of autograft and even allograft can also be used for anatomical reconstruction.

Over the past several years, successful outcomes following UCL reconstruction have been reported to be between 79% and 97%, with success defined as the patient returning to play at the same level or better than before the injury.^{5,9,17-23} Azar et al⁵ reported that 81% of patients were able to return to the



Figure 3 PNF exercise drills that include slow reversal of manual resistance. (Color version of the figure available online.)

same level of sports after UCL reconstruction, compared to 63% with primary repair. Cain et al¹⁸ reported follow-up results on 1265 UCL reconstructions, 89% of which were baseball pitchers with an average follow-up of 49 months and an 83% rate of return to the same level of competitive throwing. In a subsequent review of 256 throwing athletes who had undergone UCL reconstruction at least 10 years previously (average follow-up 12.6 years), Osbahr et al²⁰ reported that 83.5% of overhead throwers returned to participation and 98% of the group were still throwing.

Return to throwing begins with a long toss interval throwing program, no earlier than 16 weeks following UCL reconstruction. In most cases, throwing from a mound is progressed within 4-6 weeks after initiation of an interval throwing program, and return to competitive throwing is permitted at approximately 9 months after surgery. In the largest cohort of follow-up data on UCL reconstructions to date, Cain et al¹⁸ noted that the average time required to actually return to competitive throwing was 11.6 months following reconstruction. The actual rate of progression of the throwing program should be individualized to each athlete and adjusted based on symptoms, mechanics, and desired goals.

To achieve this level of success, all sources of pathology need to be addressed. Medial epicondylitis may be present. Also, concomitant valgus extension overload can result in olecranon osteophytes or loose bodies that need to be removed. If clinically present, associated ulnar neuritis, as noted previously, is addressed through nerve transposition. The other factors that may affect return to play following UCL reconstruction include the following: inadequate postsurgical stability, pain with functional activity, lack of adequate elbow and shoulder motion, incomplete rehabilitation that results in failed strength or functional examination, or all of these.

Rehabilitation Guidelines

Rehabilitation after UCL reconstruction surgery follows a sequential and progressive multiphased approach. The ultimate goal of the rehabilitation program is to return the athlete to their previous functional level as quickly and safely as possible. The rehabilitation program the authors currently use after UCL reconstruction is outlined in Table 1. The 4 phases of rehabilitation following UCL reconstruction include phase I: immediate motion phase, phase II: intermediate phase (full ROM, minimal pain, and good muscle strength), phase III: advanced strengthening phase, and phase IV: return-to-activity phase.

Phase I: Immediate Motion Phase

The goals of this phase are to minimize the effects of immobilization, reestablish pain-free range of motion, reduce pain and inflammation, and retard muscular atrophy. The athlete is often placed in a posterior splint with the elbow immobilized at 90° flexion for the first 7 days after surgery. This allows adequate healing of the soft tissue slings involved in the nerve transposition. The patient is progressed from the

posterior splint to an elbow ROM brace, which is adjusted to allow ROM of 30°-100° flexion. Motion is increased by 5° of extension and 10° of flexion thereafter to restore full ROM (0°-145°) by the end of weeks 5-6. The brace is discontinued by weeks 5-6.

Early controlled range of motion of the elbow, wrist, and forearm is performed by the rehabilitation specialist to nourish the articular cartilage and assist in the synthesis, alignment, and organization of collagen tissue.^{7,24-31} The reestablishment of either full elbow extension or preinjury motion is the primary goal of early ROM activities to minimize the occurrence of elbow flexion contractures.³¹⁻³³ Preoperative elbow motion must be carefully assessed and recorded. Patients should be asked whether they have had full elbow extension in the past 2-3 years. Postoperative ROM often is related to preoperative motion, especially with UCL reconstruction. Elbow flexion contractures can be a deleterious side effect of surgery for the overhead athlete. The elbow is predisposed to flexion contractures because of the intimate congruency of the joint articulations, the tightness of the joint capsule, and the tendency of the anterior capsule to develop adhesions after injury.²³ In addition to ROM exercises, joint mobilizations are performed as tolerated to reduce the likeliness of joint contractures. Posterior glides with oscillations are performed at end range of motion to help regain full elbow extension. Initially grade I and II mobilizations are used, with progression to aggressive mobilization techniques (grade III and IV) at end range of motion during later stages of rehabilitation, when symptoms have subsided. Joint mobilization must include the radiocapitellar and radioulnar joints.

If the patient continues to have difficulty achieving full extension using ROM and mobilization techniques, a low load-long duration stretch (also called *collagen creep* or *plastic flow stretch*) may be performed to produce a deformation or creep of the collagen tissue, resulting in tissue elongation.³³⁻³⁶ The authors have found this technique to be extremely beneficial for regaining full elbow extension. The patient lies supine with a towel or foam roll placed under the distal brachium to act as a cushion and fulcrum. Light-resistance exercise tubing is applied to the wrist and secured to the table or to a dumbbell on the ground (Fig. 1). The patient is instructed to relax as much as possible for 12-15 minutes. The patient performs this

Table 3 Advanced Thrower's 10 Exercise Program

Seated stability ball external rotation at 0° abduction [#]
Seated stability ball internal rotation at 0° abduction [#]
Seated stability ball full can [#]
Seated stability ball lateral raise to 90° [#]
Side-lying external rotation [#]
Prone T's on stability ball [#]
Prone Y's on stability ball [#]
Prone row into external rotation on stability ball [#]
Shoulder extensions on stability ball [#]
Lower trapezius isolation on stability ball [#]
Seated high row into external rotation on stability ball [#]
Seated biceps/triceps on stability ball [#]
Wrist flexion/extension and supination/pronation

[#]Sustained holds used.



Figure 4 Standing external rotation with exercise tubing at 0° abduction. (Color version of the figure available online.)

type of exercise at home periodically during the day, totaling 60 minutes of stretching per day. The resistance applied should be low enough in magnitude to enable the patient to perform the stretch for the entire duration without pain.

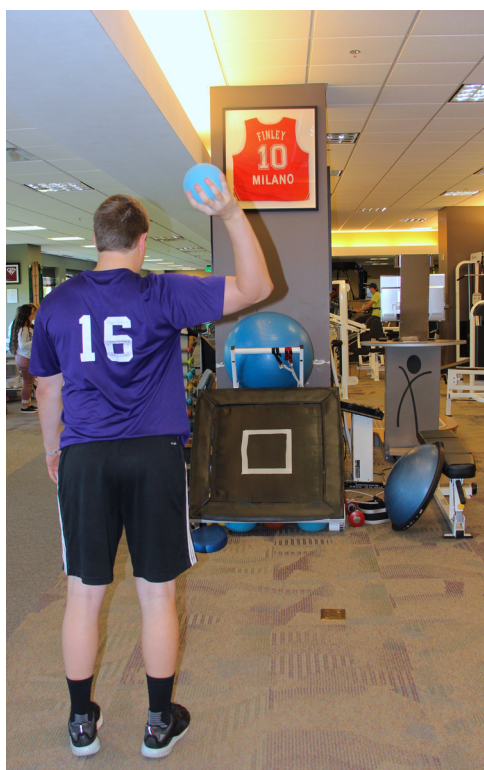


Figure 5 Plyometric drills: 1-handed baseball throw at 90° abduction. (Color version of the figure available online.)



Figure 6 Plyometric drills: 1-handed internal rotation side throws at 0° abduction. (Color version of the figure available online.)

Another goal of phase I is to reduce the patient's pain and inflammation. Grade I and grade II mobilization techniques are being used to neuromodulate pain by stimulating type I and type II articular receptors.^{37,38} Cryotherapy and high-voltage electrical stimulation may be performed as required to further assist in reducing pain and inflammation. Class IV deep-tissue LASER is used over the incision to speed healing of the incision and increase nitrous oxide levels in the healing tissue, this is to assist in graft healing. Once the acute inflammatory response has subsided, moist heat, warm whirlpool, and ultrasound may be used at the onset of treatment to prepare the tissue for stretching and improve the extensibility of the capsule and musculotendinous structures.



Figure 7 Plyometric drills: wrist flexion ball flips. (Color version of the figure available online.)



Figure 8 Plyometric drills: wrist extension with ball grips. (Color version of the figure available online.)

The early phase of rehabilitation also focuses on voluntary activation of muscle. Subpainful, submaximal isometrics are performed initially for the elbow flexor and extensor, as well as for the wrist flexor, extensor, pronator, and supinator muscle groups. Shoulder isometrics may also be performed during this phase, but care must be taken to avoid internal and external rotation exercises if these are painful. Alternating rhythmic stabilization drills for shoulder flexion or extension or horizontal abduction or adduction, shoulder internal or external rotation, and elbow flexion or extension or supination or pronation are performed to begin reestablishing proprioception and neuromuscular control of the upper extremity.

Phase II: Intermediate Phase (Full ROM, Minimal Pain, and Good Muscle Strength)

The intermediate phase is initiated when the patient exhibits full ROM, minimal pain and tenderness, and a good (4/5)

manual muscle test of the elbow flexor and extensor musculature. The emphasis in this phase includes enhancing elbow and upper extremity mobility, improving muscular strength and endurance, and reestablishing neuromuscular control of the elbow complex.

Stretching exercises are continued to maintain full elbow and wrist ROM. Mobilization techniques may be progressed to more aggressive grade III techniques as needed to apply a stretch to the capsular tissue at end range. Flexibility is progressed during this phase to focus on wrist flexion, extension, pronation, and supination. Elbow extension and forearm pronation flexibility is specifically emphasized in throwing athletes for performance. Shoulder flexibility is also maintained in athletes, with emphasis on external and internal rotation at 90° of abduction, flexion, and horizontal adduction. In particular, shoulder external rotation at 90° of abduction is emphasized because even a slight loss of external rotation may result in increased strain on the medial elbow structures during the overhead throwing motion. Internal rotation motion stretching should also be diligently performed.

We initiate light-resistance isotonic exercises at week 4 and the full thrower's 10 program by week 5 (Table 2). Progressive resistance exercises for the elbow, glenohumeral, and scapulothoracic muscles are advanced at weeks 8-9 with emphasis placed on developing dynamic stabilization of the medial elbow. Because of the anatomical orientation of the flexor carpi ulnaris and flexor digitorum superficialis overlying the UCL, isotonic and stabilization activities for these muscles may assist the UCL in stabilizing valgus stress at the medial elbow. For this reason, concentric strengthening of these muscles is also performed.

Neuromuscular control exercises are initiated in this phase to enhance the muscles' ability to control the elbow and shoulder joints during athletic activities.³⁹ These exercises include proprioceptive neuromuscular facilitation exercises with rhythmic stabilizations (Fig. 2) and slow reversal manual resistance elbow or wrist flexion drills (Fig. 3).

Table 4 The 3 "P" Program: Performance, Practice, and Play

3 "P" Program: Performance, Practice, and Play

P1: performance training	Restoration of function through sport-specific training for athletes returning to competition
P2: practice participation	Gradual progression of participation via sequentially increasing the time, intensity, and repetitions performed by the athlete during practice
P3: play	Following the successful advancement through a controlled practice progression, the athlete is evaluated for readiness to resume competition in game situations at 100% effort

Return to participation criteria

- (1) Appropriate time from injury or surgery for healing
- (2) Successful completion of a formal rehabilitation program
- (3) Full sport-specific nonpainful ROM
- (4) Excellent Stability with no pain during special tests
- (5) Strength which meets specific participation criteria
- (6) Satisfactory scores on functional tests
- (7) Sport-specific testing
- (8) Patient-reported subjective score

Table 5 Criteria to Initiate Phase I (Long Toss) Throwing

-
- Full Nonpainful ROM
 - Shoulder TROM within 5° of nonthrowing shoulder
 - Shoulder horizontal adduction 40° > on throwing shoulder
 - GIRD of < 15°
 - Elbow and wrist PROM WNL
 - Satisfactory shoulder, elbow, and wrist strength (manual muscle test, handheld dynamometer, or isokinetics)
 - ER/IR ratio: 72%-76%
 - ER/ABD ratio: 68%-73%
 - Throw shoulder IR 115% > compared to nonthrowing shoulder
 - Throw shoulder ER 95% > compared to nonthrowing shoulder
 - Elbow flex/ext 100%-115% compared to nonthrowing elbow
 - Wrist flex/ext and forearm 100%-115% compared to nonthrowing wrist/forearm
 - Satisfactory clinical examination
 - Physician clearance
 - Successfully has completed appropriate rehabilitation program
 - Completed all steps of rehabilitation
 - Satisfactory functional tests' scores
 - Prone ball drop test (throwing side 110%)
 - One arm ball throws against the wall at 90/90 (2 Lb plyoball) 30 second test with no pain 115%
 - Single-leg step-down for 30 second (8-in step)
 - Satisfactory score on the KJOC throwers' assessment
-

Table 6 Criteria to Return to Phase II (Off the Mound) Throwing

-
- Full nonpainful ROM
 - Shoulder TROM within 5° of nonthrowing shoulder
 - Shoulder horizontal adduction 40° > on throwing shoulder
 - GIRD of < 15°
 - Elbow and wrist PROM WNL
 - Satisfactory shoulder, elbow, and wrist strength (manual muscle test, handheld dynamometer, or isokinetics)
 - ER/IR ratio: 72%-76%
 - ER/ABD ratio: 68%-73%
 - Throw shoulder IR 120% > compared to nonthrowing shoulder
 - Throw shoulder ER 100% > compared to nonthrowing shoulder
 - Elbow flex/ext 110%-115% compared to nonthrowing elbow
 - Wrist flex/ext and forearm 110%-115% compared to nonthrowing wrist/forearm
 - Satisfactory clinical examination
 - Physician clearance
 - Successfully has completed appropriate rehabilitation program
 - Completed all steps of the long toss throwing program
 - Completed flat ground throwing at 70%-75% intensity
 - Satisfactory functional test scores
 - Throwing into rebounder from 20 ft with 1 Lb plyoball for 45 second at 66% intensity w/o pain, fatigue and with good mechanics
 - Prone ball drop test (throwing side 115%)
 - One arm ball throws wall throws (2 lb plyoball) 30 second test with no pain 115%
 - Single-leg step-down for 30 second (8-in step) no compensation
 - Satisfactory score on the KJOC throwers' assessment
-

Table 7 The Kerlan-Jobe Orthopaedic Clinic Shoulder and Elbow Score (KJOC)

Kerlan-Jobe Orthopaedic Clinic Shoulder & Elbow Score			
Name _____	Age _____	Sex _____	Dominant Hand (R) _____ (L) _____ (Ambidextrous)
Date of Examination _____		Sport _____	Position _____ Years Played _____
Please answer the following questions related to your history of injuries to YOUR ARM ONLY :			
	YES	NO	
1. Is your arm currently injured?	☐	☐	
2. Are you currently active in your sport?	☐	☐	
3. Have you missed game or practice time in the last year due to an injury to your shoulder or elbow?	☐	☐	
4. Have you been diagnosed with an injury to your shoulder or elbow other than a strain or sprain?	☐	☐	
If yes, what was the diagnosis? _____			
5. Have you received treatment for an injury to your shoulder or elbow?	☐	☐	
If yes, what was the treatment? (Check all that apply)			
☐ Rest ☐ Therapy ☐ Surgery (please describe): _____			
Please describe your level of competition in your current sport:			
(Use Professional Major League, Professional Minor League, Intercollegiate, High School as the choices)			
6. What is the highest level of competition you've participated at? _____			
7. What is your current level of competition? _____			
8. If your current level of competition is not the same as your highest level, do you feel it is due to an injury to your arm?	☐	☐	
Please check the ONE category only that best describes your current status:			
☐ Playing without any arm trouble ☐ Playing, but with arm trouble			
☐ Not playing due to arm trouble			
Instructions to athletes:			
The following questions concern your physical functioning during game and practice conditions. Unless otherwise specified, all questions relate to your shoulder or elbow . Please answer with an X along the horizontal line that corresponds to your current level.			
1. How difficult is it for you to get loose or warm prior to competition or practice?			
Never feel loose during games or practice Normal warm-up time			
2. How much pain do you experience in your shoulder or elbow?			
Pain at rest No pain with competition			
3. How much weakness and/or fatigue (i.e., loss of strength) do you experience in your shoulder or elbow?			
Weakness or fatigue preventing any competition No weakness, normal competition fatigue			
4. How unstable does your shoulder or elbow feel during competition?			
"Popping out" routinely No instability			
5. How much have arm problems affected your relationship with your coaches, management, and agents?			
Left team, traded or waived, lost contract or scholarship Not at all			
The following questions refer to your level of competition in your sport. Please answer with an X along the horizontal line that corresponds to your current level.			

Table 7 (continued)

6. How much have you had to change your throwing motion, serve, stroke, etc., due to your arm?	Completely changed, don't perform motion anymore No change in motion
7. How much has your velocity and/or power suffered due to your arm?	Lost all power, became linsesse or distance athlete No change in velocity/power
8. What limitation do you have in endurance in copetition due to your arm?	Significant limitation (became relief pitcher, switched to short races for example) No endurance limitation in competition
9. How much has your control (of pitches, serves, strokes, etc.) suffered due to your arm?	Unpredictable control on all pitches, serves, strokes, etc. No loss of control
10. How much do you feel your arm affects your current level of competition in your sport (i.e., is your arm holding you back from being at your full potential)?	Cannot compete, had to switch sports Desired level of competition

Phase III: Advanced Strengthening Phase

The third phase of UCL reconstruction rehabilitation involves a progression of activities to prepare the athlete for high-level stress situations and participation in sports. The goals of this phase are to gradually increase strength, power, endurance, and neuromuscular control to prepare for a gradual return to sports. Specific criteria that must be met before this phase is entered include full, nonpainful ROM; absence of pain or tenderness; and strength that is 70% of the contralateral extremity. Aggressive exercises involving eccentric and plyometric contractions are included in the advanced phase, usually weeks 9-14, and the athlete is progressed into the Advanced Thrower's 10 program (Table 3). Two-handed plyometric drills are performed at week 10, 1-handed drills at weeks 12-14.

Elbow flexion exercises are progressed to emphasize eccentric control. The biceps muscle is an important stabilizer during the follow-through phase of overhead throwing; it eccentrically controls the deceleration of the elbow, preventing pathological abutment of the olecranon within the fossa.^{19,40} Elbow flexion can be performed with elastic tubing to emphasize slow and fast concentric and eccentric contractions. Manual resistance may be applied for concentric and eccentric contractions of the elbow flexors. Aggressive strengthening exercises done with weight machines are also incorporated during this phase. These most commonly begin with bench press, seated rowing, and front latissimus dorsi pull-downs. The triceps is exercised primarily with a concentric contraction because of the acceleration activity of the triceps during the acceleration phase of throwing.

Neuromuscular control exercises are progressed to include side-lying external rotation with manual resistance.

Concentric and eccentric external rotation is performed against the clinician's resistance with the addition of rhythmic stabilizations. This manual resistance exercise may be progressed to standing external rotation with exercise tubing at 0° and finally at 90° (Fig. 4).

Plyometric drills can be an extremely beneficial form of functional exercise for training the elbow in overhead athletes.^{31,39,40} Plyometric exercises are performed using a weighted medicine ball during the later stages of phase III to train the shoulder and elbow to develop and withstand high levels of stress. Plyometric exercises initially are performed with 2 hands performing a chest pass, side-to-side throw, and overhead soccer throw. These may be progressed to include 1-handed activities such as 90/90 throws (Fig. 5), external and internal rotation throws at 0° abduction (Fig. 6), and wall dribbles. Specific plyometric drills for the forearm musculature include wrist flexion flips (Fig. 7) and extension grips (Fig. 8). Wrist flexion flips and extension grips are important components of an elbow rehabilitation program because they emphasize the forearm and hand musculature.

Phase IV: Return-to-Activity Phase

The final phase allows the athlete to progressively return to full activity and unrestricted overhead competition using an interval athletic program, be it throwing, tennis, golf, swimming, volleyball, etc.²⁷

The transition from rehabilitation back into play following an UCL reconstruction surgery follows 3 sequential steps. The first is a return to high-level performance rehabilitation and training. The second step is the resumption of participation in practice that sequentially progresses from limited, controlled activity to unrestricted practice. Finally, the third step is a return to unrestricted participation and competitive play. We refer to this program as the 3P Program: Performance, Practice, and Play (Table 4). The first 2 steps must be successfully accomplished before the athlete attempting any level of unrestricted play.

P1: Performance Training

Involves the restoration of function through sport-specific training for athletes returning to competition. This must include the ability to successfully and asymptotically perform sports-specific drills including plyometrics, end-range proprioceptive exercises, and perturbation drills. These drills differ by sport and activity but should mimic the demands and nature of the sport in a progressive and sequential manner. During this mode of training, the specific movement patterns required for asymptomatic upper quarter and elbow function are learned and integrated. A progression of challenges is provided in a controlled setting that focus on achieving and maintaining end-range stability during challenging athletic activities similar to those encountered during throwing or overhead sports.

P2: Practice Participation

Next an athlete advances into controlled participation in a practice setting, which involves the gradual progression of participation via sequentially increasing the volume dosage (including time), intensity, and repetitions performed by the athlete during practice. This progression begins at lower intensities and gradually increases from 50%-60% up to a 75% level, advancing to 80%-90% and finally 100% effort and exertion. Once 100% intensity and effort is obtained in practice, the athlete should be advanced to a practice game or scrimmage activities. In addition, in this practice, phase exposure time is also controlled, and this is to gradually improve endurance, including pitch counts for pitchers.

P3: Play

Successfully advancing functional activity beyond performance training and back into participation in practice and then competitive play requires more than just a decision based on subjective criteria, such as how the athlete says their elbow feels. The complex nature of progressing an athlete back into unrestricted participation following any shoulder or elbow injury, including UCL reconstruction surgery, requires assessing and measuring the key functional elements necessary for symptom-free athletic performance of the elbow and upper extremity. The athlete must also demonstrate sufficient confidence in the affected extremity to successfully return to sport without any fears or limitations. The criteria we evaluate to assist in determining when an athlete can return to practice and then unrestricted competition, include the following: (1) appropriate time from injury or surgery based on tissue healing, (2) successful completion of a formal rehabilitation program, (3) full sport-specific nonpainful ROM, (4) excellent stability with no pain during special tests, (5) strength which meets specific participation criteria, (6) satisfactory functional scoring, and (7) patient-reported subjective score (Table 4).

These tests are designed to determine activity readiness before the introduction of demanding functional athletic

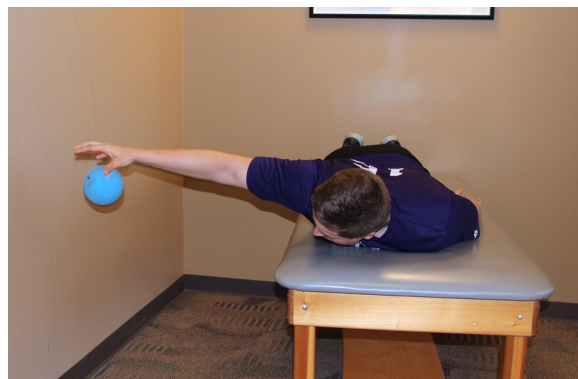


Figure 9 Ball drop test: patient is in prone position on table with 2 Lb plyoball in hand performs ball drops and catches for 30 second with the shoulder abducted to 90° and elbow extended. (Color version of the figure available online.)

Table 8 Phase I Long Toss Interval Throwing Program

The interval throwing program (ITP) is designed to gradually return motion, strength, and confidence in the throwing arm after injury or surgery by slowly progressing through graduated throwing distances. The ITP is initiated upon clearance by the athlete's physician to resume throwing, and performed under the supervision of the rehabilitation team (physician, physical therapist, and athletic trainer).

The program is set up to minimize the chance of reinjury and emphasize prethrowing warm-up and stretching. In development of the interval throwing program, the following factors are considered most important.

- (1) The act of throwing the baseball involves the transfer of energy from the feet through the legs, pelvis, trunk, and out the shoulder through the elbow and hand. Therefore, any return to throwing after injury must include attention to the entire body.
- (2) The chance for reinjury is lessened by a graduate progression of interval throwing.
- (3) Proper warm-up is essential.
- (4) Most injuries occur as the result of fatigue.
- (5) Proper throwing mechanics lessen the incidence of reinjury.
- (6) Baseline requirements for throwing include:

- Pain-free range of motion
- Adequate muscle power
- Adequate muscle resistance to fatigue

Because there is an individual variability in all throwing athletes, there is no set timetable for completion of the program. Most athletes, by nature, are highly competitive individuals and wish to return to competition at the earliest possible moment. Although this is a necessary quality of all athletes, the proper channeling of the athlete's energies into a rigidly controlled throwing program is essential to lessen the chance of reinjury during the rehabilitation period. The athlete may have the tendency to want to increase the intensity of the throwing program. This will increase the incidence of reinjury and may greatly retard the rehabilitation process. It is recommended to follow the program rigidly as this will be the safest route to return to competition.

During the recovery process, the athlete will probably experience soreness and a dull, diffuse aching sensation in the muscles and tendons. If the athlete experiences sharp pain, particularly in the joint, stop all throwing activity until this pain ceases. If continued pain, contact your physician.

Weight training: The athlete should supplement the ITP with a high repetition, low-weight exercise program. Strengthening should address a good balance between anterior and posterior musculature so that the shoulder will not be predisposed to injury. Special emphasis must be given to posterior rotator cuff musculature for any strengthening program. Weight training will not increase throwing velocity, but will increase the resistance of the arm to fatigue and injury. Weight training should be done the same day as you throw; however, it should be after your throwing is completed, using the day in between for flexibility exercises and a recovery period. A weight training pattern or routine should be stressed at this point as a "maintenance program." This pattern can and should accompany the athlete into and throughout the season as a deterrent to further injury. It must be stressed that weight training is of no benefit unless accompanied by a sound flexibility program.

Individual variability: The ITP is designed so that each level is achieved without pain or complications before the next level is started. This sets up a progression that a goal is achieved before advancement instead of advancing to a specific timeframe. Because of this design, the ITP may be used for different levels of skills and abilities from those in high school to professional levels. The reasons for being in the ITP will vary from person to person. Example: one athlete may wish to use alternate days throwing with or without using weights in between; another athlete may have to throw every third or fourth day due to pain or swelling. "Listen to your body—it will tell you when to slow down." Again, completion of the steps of the ITP will vary from person to person. There is no set timetable in terms of days to completion.

Warm-up: Jogging increases blood flow to the muscles and joints, thus increasing their flexibility and decreasing the chance of reinjury. As the amount of warm-up will vary from person to person, the athlete should jog until developing a light sweat, then progress to the stretching phase.

Stretching: As throwing involves all muscles in the body, all muscle groups should be stretched before throwing. This should be done in a systematic fashion beginning with the legs and including the trunk, back, neck, and arms. Continue with capsular stretches and L-bar range of motion exercises.

Throwing mechanics: A critical aspect of the ITP is maintenance of proper throwing mechanics throughout the advancement. The use of the Crow-Hop method simulates the throwing act, allowing emphasis of the proper body mechanics. This throwing method should be adopted from the set of the ITP. Throwing flat footed encourages improper body mechanics, placing increased stress on the throwing arm and, therefore, predisposing the arm to reinjury. The pitching coach and sports biomechanicianist (if available) may be valuable allies to the rehabilitation team with their knowledge of throwing mechanics.

Table 8 (continued)

Components of the Crow-Hop method are first a hop, then a skip, followed by the throw. The velocity of the throw is determined by the distance, whereas the ball should have only enough momentum to travel each designed distance. Again, emphasis should be placed upon proper throwing mechanics when the athlete beings phase 2: "Throwing Off the Mound" or from the athlete's respective position, to decrease the chance of reinjury.

Throwing: Using the Crow-Hop method, the athlete should begin warm-up throws at a comfortable distance (approximately 30-45 ft) and then progress to the distance indicated for that phase (Table 1). The program consists of throwing at each step 2-3 times without pain or symptoms before progressing to the next step. The object of each phase is for the athlete to be able to throw the ball without pain the specified number of feet (45, 60, 90, 120, 150, and 180 ft), 75 times at each distance. After the athlete can throw at the prescribed distance without pain, they will be ready for throwing from flat ground 60 ft, 6 in. in the normal pitching mechanics or return to their respective position (step 14). At this point, full strength and confidence should be restored in the athlete's arm. It is important to stress the Crow-Hop method and proper mechanics with each throw. Just as the advancement to this point has been gradual and progressive, the return to unrestricted throwing must follow the same principles. A pitcher should first throw only fastballs at 50%, progressing to 75% and 100%. At this time, he may start more stressful pitches such as breaking balls. The position player should simulate a game situation, again progressing at 50%-75% and then 100%. Once again, if an athlete has increased pain, particularly at the joint, the throwing program should be backed off and readvanced as tolerated, under the direction of the rehabilitation team.

Batting: Depending on the type of injury that the athlete has, the time of return to batting should be determined by the physician. It should be noted that stress placed upon the arm in the batting motion is very different from the throwing motion. Return to unrestricted use of the bat should also follow the same progression guidelines as seen in the training program. Begin with dry swings progressing to hitting off the tee, then soft toss, and finally live pitching.

Summary: In using the interval throwing program (ITP) in conjunction with a structured rehabilitation program, the athlete should be able to return to full competition status, minimizing any chance of reinjury. The program and its progression should be modified to meet the specific needs of each individual athlete. A comprehensive program consisting of a maintenance strength and flexibility program, appropriate warm-up and cool-down procedures, proper pitching mechanics, and progressive throwing and batting will assist the baseball player in returning safely to competition.

45' Phase		60' Phase		90' Phase		120' Phase	
Step1:	Warm-up throwing 45' (25 Throws) Rest 5-10 minute Warm-up throwing 45' (25 throws)	Step 3:	Warm-up throwing 60' (25 throws) Rest 5-10 minute Warm-up throwing 60' (25 throws)	Step 5:	Warm-up throwing 90' (25 throws) Rest 5-10 minute Warm-up throwing 90' (25 throws)	Step 7:	Warm-up throwing 120' (25 throws) Rest 5-10 minute Warm-up throwing 120' (25 throws)
Step 2:	Warm-up throwing 45' (25 throws) Rest 5-10 minute Warm-up throwing 45' (25 throws) Rest 5-10 minute Warm-up throwing 45' (25 throws)	Step 4:	Warm-up throwing 60' (25 throws) Rest 5-10 minute Warm-up throwing 60' (25 throws) Rest 5-10 minute Warm-up throwing 60' (25 throws)	Step 6:	Warm-up throwing 90' (25 throws) Rest 5-10 minute Warm-up throwing 90' (25 throws) Rest 5-10 minute Warm-up throwing 90' (25 throws)	Step 8:	Warm-up throwing 120' (25 throws) Rest 5-10 minute Warm-up throwing 120' (25 throws) Rest 5-10 minute Warm-up throwing 20' (25 throws)
150' Phase		180' Phase					
Step 9:	Warm-up throwing 150' (25 throws) Rest 5-10 minute Warm-up throwing 150' (25 throws)	Step 11:	Warm-up throwing 180' (25 throws) Rest 5-10 minute Warm-up throwing 180' (25 throws)	Step 13:	Warm-up throwing 180' (25 throws) Rest 5-10 minute Warm-up throwing 180' (25 throws)	Throwing Program should be performed every other day, unless otherwise specified by your physician or rehabilitation specialist	
Step 10:	Warm-up throwing 150' (25 throws) Rest 5-10 minute Warm-up throwing 150' (25 throws) Rest 5-10 minute Warm-up throwing 150' (25 throws)	Step 12:	Warm-up throwing 180' (25 throws) Rest 5-10 minute Warm-up throwing 180' (25 throws) Rest 5-10 minute Warm-up throwing 180' (25 throws)	Step 14:	Begin throwing off the mound or return to respective position		
				Perform each step __ times before progressing to next step			

Table 8 (continued)

Flat ground throwing	Flat throwing
(A) Warm-up throwing	(A) Warm-up throwing
(B) Throw 60 ft (10-15 throws)	(B) Throw 60 ft (10-15 throws)
(C) Throw 90 ft (10 throws)	(C) Throw 90 ft (10 throws)
(D) Throw 120 ft (10 throws)	(D) Throw 120 ft (10 throws)
(E) Throw 60 ft (flat ground) using pitching mechanics (20-30 throws)	(E) Throw 60 ft (flat ground) using pitching mechanics (20-30 throws)
	(F) Throw 60-90 ft (10-15 throws)
	(G) Throw 60 ft (flat ground) using pitching mechanics (20 throws)

practice or competitive participation, to reduce the risk of reinjury, or the development of other related issues; as well as promote psychological confidence in the involved elbow and throwing arm. This program rank-orders the relative demand of functional activities required for athletic participation and guides advancement back to unlimited activity via defined criteria to determine readiness for resuming practice and then unrestricted participation following elbow injury (Tables 5 and 6).

Patient-Reported Outcome Tools

The American Shoulder and Elbow Surgeon's (ASES) Shoulder Score assessment tool is administered to assist in determining the athletes' self-assessment of the functional status and confidence in their throwing arm. Patients' self-reporting scoring can be performed with the Kerlan-Jobe Orthopaedic Clinic Shoulder and Elbow Score (KJOC).¹⁵ This evaluation form is beneficial for the overhead athlete who has been treated for elbow and shoulder injury or instability (Table 7).

Range of Motion

Before the return-to-practice participation (P2), the overhead athlete should have full, nonpainful range of motion. For the overhead athlete, full range of motion is shoulder total rotational range of motion within 5° of the nonthrowing shoulder for both IR and ER motions.⁴¹ The athlete should also exhibit horizontal shoulder adduction >40° on the throwing shoulder. Glenohumeral joint internal rotation deficit as compared to the nondominant shoulder should be <15° and asymptomatic.⁴² Elbow and wrist range of motion has been studied by the authors during spring training physicals for the past 15 years. We have found that the average PROM in a professional baseball pitcher is seen to be 0°-147°. Wright et al⁴³ reported on 33 professional pitchers and stated that the throwers' elbow lacked 7° of extension and 5.5° of flexion, with total motion (extension + flexion = total motion) to be lacking by 13°. These investigators' results are consistent with ours. The thrower's elbow often has limited elbow extension compared to the nonthrowing side. We have found that the pitcher's mean supination to be 93° and pronation 98°.

Isokinetic Test and Hand-Held Dynamometry

Isokinetic testing or handheld dynamometer assessment of the shoulder internal and external rotators and elbow flexors and extensors is also conducted to assess throwing arm strength and power. It is important to closely evaluate the specific parameters to ensure that the athletes exhibit appropriate strength and power relative to the ratio between their muscle groups, their ability to generate torque in relation to their body weight, as well as bilateral comparison of strength.^{44,45}

Ball Drop Test

We have developed a specific test to evaluate throwing arm endurance, willingness to move quickly, and dynamic stability called the ball drop test. It is performed in the prone position with the arm abducted at 90° with the elbow extended, using a 1 kg (2-pound) weighted ball with the arm being tested completely off the plinth as a measure of dynamic stability of the shoulder (Fig. 9). The test is performed for a bout of 30 seconds counting the number of releases and catches and then compared involved to uninvolved side for a performance percentage. We have determined a satisfactory score of >110% on the dominant extremity.

Wall Throws Test at 90°/90°

We have developed another test for the overhead throwing athlete to assess endurance, strength, proper mechanics, and proprioception. The athlete stands in a doorway and throws a 2 Lb plyoball against the wall at 90° of abduction for 30 second. We calculate the number of throws on both the dominant and nondominant shoulder. Based on data collection performed on overhead throwing athletes, the bilateral difference is 112% or greater. In addition, higher-level athletes (professional baseball players) exhibit a higher difference than high school or collegiate athletes.

Single-Leg Step-Down Test

The single-leg step-down test is a test used to assess the strength and stability of the lower extremities. The time

Table 9 Phase II Off The Mound Interval Throwing Program

After the completion of phase I of the interval throwing program (ITP) and the athlete can throw to the prescribed distance without pain, the athlete then will be ready for throwing off the mound or return to their respective position. At this point, full strength and confidence should be restored in the athlete's arm. Just as the advancement to this point has been gradual and progressive, the return to unrestricted throwing must follow the same principles. A pitcher should first throw only fast ball at 50%, progressing to 75%, and then 100%. At this time, the athlete may start more stressful pitches such as breaking balls. The position player should simulate a game situation, again progressing at 50%-100%. Once again, if an athlete has increased pain, particularly at the joint, the throwing program should be backed off and re-advanced as tolerated, under the direction of the rehabilitation team.

Summary: In using the interval throwing program (ITP) in conjunction with a structured rehabilitation program, the athlete should be able to return to full competition status, minimizing any chance of reinjury. The program and its progression should be modified to meet the specific needs of each individual athlete. A comprehensive program consisting of a maintenance strength and flexibility program, appropriate warm-up and cool-down procedures, proper pitching mechanics, and progressive throwing and batting will assist the baseball player in returning safely to competition.

Stage 1: fastballs only

Step 1	Interval throwing 15 throws off mound 50%	(Use interval throwing to 120' Phase as warm-up)
Step 2	Interval throwing 30 throws off mound 50%	
Step 3	Interval throwing 45 throws off mound 50%	All throwing off the mound should be done in the presence of your pitching coach to stress proper throwing mechanics
Step 4	Interval throwing 60 throws off mound 50%	
Step 5	Interval throwing 70 throws off mound 50%	(Use speed gun to aid in effort control)
Step 6	45 throws off mound 50% 30 throws off mound 75%	
Step 7	30 throws off mound 50% 45 throws off mound 75%	
Step 8	65 throws off mound 75% 10 throws off mound 50%	

Stage 2: fastballs only

Step 9	60 throws off mound 75% 15 throws in Batting Practice
Step 10	50-60 throws off mound 75% 30 throws in Batting Practice
Step 11	45-50 throws off mound 75% 45 throws in batting practice

Stage 3

Step 12	30 throws off mound 75% warm-up 15 throws off mound 50% breaking balls 45-60 throws in Batting Practice (fastball only)
Step 13	30 throws off mound 75% 30 Breaking Balls 75% 30 throws in Batting Practice
Step 14	30 throws off mound 75% 60-90 throws in Batting Practice (Gradually increase breaking balls)
Step 15	Simulated game: progressing by 15 throws per workout (pitch count)

allotted for this test is 30 seconds; to achieve a satisfactory score, the athlete must be able to perform 10 repetitions. The clinician must keep note of the pelvis and trunk positioning, specifically the presence of a contralateral drop and ipsilateral lean as compensatory patterns for weakness. Although this test is not a test of the elbow specifically, weakness of the lower extremities in single-leg stance can contribute to altered mechanics in the throwing motion potentially making the elbow more susceptible to reinjury.

Upon achieving the previously mentioned criteria for returning to sports, the authors begin a formal interval sports program as described by Reinold et al.⁴⁶ For the overhead thrower, the authors initiate a long toss interval throwing program beginning at 45 ft and gradually progressing to 120 or 180 ft (player and position dependent) (Table 8).⁴⁶ Throwing should be performed without pain or a significant increase in symptoms. The authors believe it is important for the overhead patient or athlete to perform stretching and an abbreviated strengthening program before and after performing the interval sports program. Typically, overhead throwers warm-up, stretch, and perform 1 set of their exercise program before throwing, followed by 2 additional sets of exercises following throwing.²⁷ This not only provides an adequate warm-up but also ensures maintenance of necessary range of motion and flexibility of the shoulder joint. The next day, the thrower exercises the scapular muscles and external rotators and performs a core stabilization program.

After completing a long toss program, pitchers progress to phase II of the throwing program, throwing off of a mound (Table 9).²⁷ In phase II, the number of throws, intensity, and type of pitch are progressed to gradually increase stress on the elbow and shoulder joints.²⁷ Generally, the pitcher begins at 50% intensity and gradually progresses to 75%, 90%, and 100% over 4-6 weeks. Breaking balls are initiated once the pitcher can throw 40-50 pitches with at least 80% intensity with no symptoms.

Summary

The elbow is a complex joint, important for positioning the hand and space and in the specialized functional demands of the overhead athletics. The UCL is a frequently injured structure in overhead throwing athletes that requires specific interventions and a dedicated rehabilitation program to obtain a satisfactory clinical result and return the throwing athlete to their previous level of function. Clearly, the anatomy and biomechanics of the elbow guide our understanding of the pathophysiology of elbow injury and help direct the treatment of elbow conditions. However, the elbow's response to trauma, even iatrogenic trauma, can be tricky, because stiffness or loss of motion is a common problem and can be a challenge in restoring function and regaining symptom-free athletic capability. We have outlined a rehabilitation program following UCL reconstruction and performance progression assessment testing program that provides the clinician with a useful set of tools to objectively assist and guide the determination regarding when an overhead throwing athlete can safely progress

back to practice and then return to unrestricted athletic activities. In addition, the results of these tests provide information regarding the patient's status in the rehabilitation program and may also provide incentive for patients who require additional strength and neuromuscular retraining. The testing battery purposely incorporates a subjective analysis, conducted by the clinician, regarding the athlete's functional ability. A lack of confidence or any compensation strategies used during these tasks indicates an athlete who requires continued training and counseling before returning to unrestricted athletics.

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